

Spring 1986

Problem 1 (Sp86). Let $e = (a, b, c)$ be a unit vector in $\mathbb{R}^3$ and let T be the linear transformation on $\mathbb{R}^3$ of rotation by 180° about e . Find the matrix for T with respect to the standard basis.

Problem 2 (Sp86). Let f be a continuous real valued function on $\mathbb{R}$ such that

$$f(x) = f(x+1) = f\left(x + \sqrt{2}\right)$$

for all x . Prove that f is constant.

Problem 3 (Sp86). Let C be a simple closed contour enclosing the points $0, 1, 2, \dots, k$ in the complex plane, with positive orientation. Evaluate the integrals

$$I_k = \int_C \frac{dz}{z(z-1)\cdots(z-k)} \quad k = 0, 1, \dots$$

$$J_k = \int_C \frac{(z-1)\cdots(z-k)}{z} dz \quad k = 0, 1, \dots$$

Problem 4 (Sp86). Let f be a positive differentiable function on $(0, \infty)$. Prove that

$$\lim_{\delta \rightarrow 0} \left(\frac{f(x+\delta x)}{f(x)} \right)^{1/\delta}$$

exists (finitely) and is nonzero for each x .

Problem 5 (Sp86). Prove that there exists only one automorphism of the field of real numbers; namely the identity automorphism.

Problem 6 (Sp86). Let V be a finite dimensional vector space and A and B two linear transformations of V into itself such that $A^2 = B^2 = 0$ and $AB + BA = I$.

1. Prove that $\ker A = A \ker B$, $\ker B = B \ker A$, and $V = \ker A \oplus \ker B$.
2. Prove that the dimension of V is even.
3. Prove that if the dimension of V is 2, then V has a basis with respect to which A and B are represented by the matrices

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Problem 7 (Sp86). For λ a real number, find all solutions of the integral equations

$$\varphi(x) = e^x + \lambda \int_0^x e^{(x-y)} \varphi(y) \, dy \quad 0 \leq x \leq 1$$

$$\psi(x) = e^x + \lambda \int_0^1 e^{(x-y)} \psi(y) \, dy \quad 0 \leq x \leq 1$$

where $\varphi, \psi : [0, 1] \rightarrow \mathbb{R}$ are continuous with continuous derivatives on $(0, 1)$.

Problem 8 (Sp86). Let the 3×3 matrix function A be defined on the complex plane by

$$A(z) = \begin{pmatrix} 4z^2 & 1 & -1 \\ -1 & 2z^2 & 0 \\ 3 & 0 & 1 \end{pmatrix}$$

How many distinct values of z are there such that $|z| < 1$ and $A(z)$ is not invertible?

Problem 9 (Sp86). Let $\mathbb{Z}^2$ be the group of lattice points in the plane, i.e., ordered pairs of integers, with coordinatewise addition as the group operation. Let H_1 be the subgroup generated by the two elements $(1, 2)$ and $(4, 1)$, and H_2 the subgroup generated by the two elements $(3, 2)$ and $(1, 3)$. Are the quotient groups $G_1 = \mathbb{Z}^2/H_1$ and $G_2 = \mathbb{Z}^2/H_2$ isomorphic?

Problem 10 (Sp86). Suppose addition and multiplication are defined on $\mathbb{C}^n$, complex n -space, coordinatewise, making $\mathbb{C}^n$ into a ring. Find all ring homomorphisms of $\mathbb{C}^n$ onto $\mathbb{C}$.

Problem 11 (Sp86). Let the complex valued functions $f_n, n \in \mathbb{Z}$, be defined on $\mathbb{R}$ by

$$f_n(x) = \frac{(x-i)^n}{\sqrt{\pi}(x+i)^{n+1}}$$

Prove that these functions are *orthonormal*; that is,

$$\int_{-\infty}^{\infty} f_m(x) \overline{f_n(x)} \, dx = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$$

Problem 12 (Sp86, Sp24). Let f be a real valued continuous function on $\mathbb{R}$ satisfying the *mean value inequality* below:

$$f(x) \leq \frac{1}{2h} \int_{x-h}^{x+h} f(y) \, dy \quad x \in \mathbb{R} \quad h > 0$$

Prove:

1. The maximum of f on any closed interval is assumed at one of the endpoints.
2. f is convex.

Problem 13 (Sp86). Let S be a nonempty commuting set of $n \times n$ complex matrices ($n \geq 1$). Prove that the members of S have a common eigenvector.

Problem 14 (Sp86, Sp94, Sp96, Fa98). Let K be a compact subset of $\mathbb{R}^n$ and $\{B_j\}$ a sequence of open balls that covers K . Prove that there is a positive number ε such that each ε -ball centered at a point of K is contained in one of the balls B_j .

Problem 15 (Sp86). Let T be an isometry of $\mathbb{R}^2$ into itself. Prove that T can be represented as $T(x) = a + U(x)$, where a is a vector in $\mathbb{R}^2$ and U is an orthogonal linear transformation.

Problem 16 (Sp86). Determine whether the rings

$$\mathbb{F}_p[x]/\langle x^2 - 2 \rangle \quad \text{and} \quad \mathbb{F}_p[x]/\langle x^2 - 3 \rangle$$

are isomorphic in each of the cases $p = 2, 5$, and 11 .

Problem 17 (Sp86). Let V be a finite dimensional vector space over $\mathbb{C}$ of $\mathcal{C}^\infty$ complex valued functions on $\mathbb{R}$, the linear operations being defined point-wise. Prove that if V is closed under differentiation, i.e., $f'(x)$ belongs to V whenever $f(x)$ does, then V is closed under translations, i.e., $f(x + a)$ belongs to V whenever $f(x)$ does, for all real numbers a .

Problem 18 (Sp86). Let $f, g_1, g_2, \dots$ be entire functions. Assume that

- $|g_n^{(k)}(0)| \leq |f^{(k)}(0)|$ for all n and k ;
- $\lim_{n \rightarrow \infty} g_n^{(k)}(0)$ exists for all k .

Prove that the sequence $\{g_n\}$ converges uniformly on compact sets and that its limit is an entire function.

Problem 19 (Sp86). Prove that the additive group of $\mathbb{Q}$, the rational number field, is not finitely generated.

Note: See also ??????.

Problem 20 (Sp86). Evaluate $\int_{|z|=1} \frac{1}{(e^{2\pi z} + 1)^2} dz$ where the integral is taken in counterclockwise direction.